

Institute of Actuaries of India

Subject CT1 – Financial Mathematics

March 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) a) Certificate of deposit:**

- Short term investments
- A CoD is a certificate stating that some money has been deposited
- Issued by banks and building societies
- Interest is payable on maturity
- Terms to maturity are usually in the range of 28 days to 6 months.
- Degree of security and marketability depends on the issuer.
- There is an active secondary market in certificate of deposits

[2]

b) Government Index linked bond:

- Considered to be very secure form of investments
- Bonds issued and payments made by the Government
- Coupon and redemption payments are linked an index for e.g. RPI
- Generally bond provides inflation protection

[2]

ii) Characteristics of property investment:

- Large unit sizes
- Each property is unique hence difficult to value
- The actual value obtained on sale is uncertain
- Buying and selling expenses are higher
- Net rental income may be reduced due to maintenance expenses
- There may be period when property is unoccupied, loss of rental income
- Marketability is poor
- The running yield from property investment would be higher

[4]

iii) $(1 - (90/365) * 0.08) = (1+i)^{-(90/365)}$

$$0.980274 = (1+i)^{-(90/365)}$$

$$i = 8.41\%$$

[2]

[10 marks]

Solution 2:

$$\begin{aligned}
 \text{i) a) } {}_5|a_{32} &= a_{37} - a_5 \quad @ 4.0\% \\
 &= 19.1425 - 4.4518 \\
 &= \mathbf{14.691}
 \end{aligned}$$

[2]

$$\begin{aligned}
 \text{b) } {}_4|(Ia)_{15} & \\
 &= v^4(Ia)_{15} \\
 &= 0.854804 \times ({}_{15}\ddot{a}_{15} - 15v^{15}) / 0.04 \\
 &= 0.854804 \times (11.5631 - 8.32896) / 0.04 \\
 &= \mathbf{69.11}
 \end{aligned}$$

[2]

$$\begin{aligned}
 \text{ii) Since } \ddot{a}_{2n} &= \ddot{a}_n + v^n \ddot{a}_n = \ddot{a}_n (1 + v^n) \\
 (1 + v^n) &= 10.934563 / 7.029584 \\
 \text{So } v^n &= 0.555506 \\
 \text{Now } \ddot{a}_n &= (1 - v^n) / d = (1 - 0.555506) / d = 7.029584 \\
 d &= .063232 \text{ and } i = d / (1 - d), i = 6.75\% \\
 n \log v &= \log (0.555506) \\
 n &= \log(0.555506) / \log(1 - 0.063232) \\
 n &= \mathbf{9}
 \end{aligned}$$

[3]

[7 marks]

Solution 3:

Assuming an inflation rate of j , the equation of value :

$$(50000 + 500) / (1 + j) \times a_{15} = 5,00,000, \quad \text{calculated @ } i' = ((1.08) / (1 + j)) - 1$$

This can be written as :

$$(1.08) / (1 + j) \times a_{15} = 1.08 \times 5,00,000 / 50,500$$

$$(1.08) / (1 + j) \times a_{15} = 10.693$$

Or

$$\ddot{a}_{15} = 10.693 \text{ or } a_{14} = 9.693 \quad \text{calculated @ } i' = ((1.08) / (1 + j)) - 1$$

$$i' = 5\%, \text{ LHS} = 9.89$$

$$i' = 5.5\%, \text{ LHS} = 9.58$$

By interpolation, $i' = 5.3177\%$

$$\frac{1.08}{1 + j} - 1 = .053177$$

$$1 + j$$

$$j = 2.55\%$$

[5 marks]

Solution 4:

i) Let the loan installment be Rs P

$$P \times a_{\overline{25}|} = 25,000,00 \quad @10\%$$

$$P \times 9.077 = 25,00,000 = P = 2,75,421.40$$

[2]

ii) a) Tenth Year

$$\text{Loan outstanding after 9th year} = P \times a_{\overline{16}|} = 2,75,421.40 \times 7.8237$$

$$\text{Interest payment during tenth year} = 0.1 \times 2,75,421.40 \times 7.8237 = \text{Rs } 2,15,481.4$$

$$\text{Capital payment during tenth year} = 2,75,421.40 - 2,15,481.4 = \text{Rs } 59,939.96$$

[2]

b) Fifteenth Year

$$\text{Loan outstanding after 14th year} = P \times a_{\overline{11}|} = 2,75,421.40 \times 6.4951$$

$$\text{Interest payment during fifteen year} = 0.1 \times 2,75,421.40 \times 6.4951 = \text{Rs } 1,78,888.95$$

$$\text{Capital payment during fifteen year} = 2,75,421.40 - 1,78,888.95 = \text{Rs } 96,532.45$$

[2]

iii) Let after t^{th} repayment the loan outstanding will be first less than 12,00,000

Then we have

$$2,75,421.40 \times a_{\overline{25-t}|} < 12,00,000$$

$$a_{\overline{25-t}|} < 4.35696$$

$$(1 - (1.1)^{-(t-25)}) < 0.435696$$

$$(t-25) \log(1.1) > \log(0.5643)$$

$$(t-25) > -6$$

$$t > 19 \text{ yrs}$$

After 19th repayment the capital outstanding will be first less than Rs 12,00,000

[3]

iv) Let the capital content exceeds the interest content in the t^{th} installment

$$\text{Interest payment in } t^{\text{th}} \text{ year} = 0.1 \times 2,75,421.40 \times a_{\overline{25-(t-1)}|}$$

$$\text{Capital payment in } t^{\text{th}} \text{ year} = 2,75,421.40 \times (1 - 0.1 \times a_{\overline{25-(t-1)}|})$$

We have:

$$\Rightarrow 0.1 \times 2,75,421.40 \times a_{\overline{25-(t-1)}|} < 2,75,421.40 \times (1 - 0.1 \times a_{\overline{25-(t-1)}|})$$

$$\Rightarrow (1 - v^{(26-t)}) < v^{(26-t)}$$

$$= v^{(26-t)} > 0.50 = (1.1)^{-(t-26)} > 0.50$$

$$= (t-26) > \log(0.50) / \log(1.1)$$

$$= (t-26) > -7.27$$

$$= t > 18.74$$

In the 19th year of payment the capital content will first exceeds the interest content.

[4]

[13 marks]

Solution 5:*(Working in units of Rs crores)*

i) Until the initial loan is repaid, the company's balance just after the transaction at time t years is:

$$-100 (1+i)^t - 80 \times S_{t|} + 120 \times S_{t|} = 0 @ 15 \%$$

$$-100 (1+i)^t + 40 \times S_{t|} = 0$$

$$40 \times a_{t|} = 100$$

$$(1 - (1.15)^{-t}) / 0.15 = 2.50$$

$$(1.15)^{-t} = 0.625$$

$$-t \log(1.15) = \log 0.625$$

$$t = 3.36$$

The bank loan will be paid off after 4th year.

[4]

ii) The balance at time 4 years, just after repaying the bank loan is

$$-100 (1+i)^4 + 40 \times S_{4|} @ 15\%$$

$$= -174.90 + 40 \times 4.9933$$

$$= -174.90 + 199.735 = 24.835$$

The balance in the company's bank account in thirteen years, when the project ends, is thus:

$$= 24.835 \times (1+i)^9 + (120 \times S_{6|} - 80 \times S_{6|}) \times (1+i)^3 - 80 S_{3|} @ 12\%$$

$$= 68.869 + 40 \times ((1.12)^6 - 1) / .12 \times 1.12^3 - 80 \times ((1.12)^3 - 1) / .12$$

$$= 68.869 + 40 \times 11.4012 - 80 \times 3.3744$$

$$= 254.97$$

[6]

iii) **Discounted Payback period** – The DPP for an investment project is the earliest time after the start of the project when the accumulated value of past cashflows (positive and negative), calculated using the borrowing rate, becomes greater or equal to zero.

i.e. Acc. Profit at discounted payback period ≥ 0

Or

The DPP is the smallest time t for which the PV of the returns up to time t exceeds the PV of the cost up to time t .

Payback period - The payback period is same as the DPP, except that the PV calculation is carried out using an interest rate of zero. In other words, it is the earliest time for which the monetary value of the return exceeds monetary value of costs.

Not considered as good measures:

- Unlike NPV, neither DPP nor the PP gives an indication how profitable the project is, as they ignore cashflows after the accumulated value of zero is reached.
- There may not be one unique time when the balance in the investors account changes from positive to negative
- The PP can be misleading, as it does not take into account the time value of money.

[3]

[13marks]**Solution 6:****i) (Working in '000s)**

Date*	Cashflows	Description
01/01/2016	-92	Purchase price
30/06/2016	$= 2.5 \times (102/100) = 2.55$	Coupon
31/12/2016	$= 2.5 \times (104/100) = 2.60$	Coupon
30/06/2017	$= 2.5 \times (108/100) = 2.70$	Coupon
31/12/2017	$= 2.5 \times (111/100) = 2.775$	Coupon
30/06/2018	$= 2.5 \times (113/100) = 2.825$	Coupon
31/12/2018	$= 2.5 \times (115/100) = 2.875$	Coupon
31/12/2018	$= 100 \times (115/100) = 115$	Redemption amount

(*Inflation is assumed to remain constant over a month. Thus inflation is assumed to remain constant between 01st June and 30th June and between 01st December and 31st December.)

[4]

ii) a) Capital gain tax paid by investor :

$$0.30 \times (115 - 92 \times (118/102)) = 2.57$$

[2]

b) Net effective yield :

$$92 = 0.80 \times (2.55 \times v^{0.5} + 2.60 \times v^1 + 2.70 \times v^{1.5} + 2.775 \times v^{2.0} + 2.825 \times v^{2.5} + 2.875 \times v^3) + (115 - 2.57) \times v^3$$

At $i = 11\%$, LHS = 93.09

At $i = 12\%$, LHS = 90.74

By interpolation, $i = 11.47\%$

Net effective yield to investor is 11.47%

[4]

[10marks]

Solution 7:

Option 1:

The present value of the shares in one month's time is:

$$= [8 \times 1.04^2 \times v + 8 \times 1.04^3 \times v^2 + \dots]$$

So the present value one month earlier is

$$= v^{1/12} [8 \times 1.04^2 \times v + 8 \times 1.04^3 \times v^2 + \dots]$$

$$= v^{1/12} \times 8 \times 1.04 [1.04 \times v + 1.04^2 \times v^2 + \dots]$$

Hence the return can be calculated as

$$120 = v^{1/12} \times 8.32 \times a_{\infty} \text{ (at } i' \text{ where } i' = \frac{1+i}{1.04} - 1)$$

By trial and error $i = 11\%$

Option 2:

$$8000 = 600 a_{10} + 10000 \times v^{10}$$

By trial and error $i = 9\%$

Option 3:

Price of fixed interest security per 100 nominal is

$$= 6 [a_{27\%} + (v^2 a_{2\%})_{8\%} + (v^4 a_{2\%})_{9\%} + (v^6 a_{2\%})_{10\%} + (v^8 a_{2\%})_{11\%}] + 110 v^{10}_{11\%}$$

$$= 6 [1.8080 + (0.8573 \times 1.7833) + (0.7084 \times 1.7591) + (0.5644 \times 1.7355) + (0.4399 \times 1.7125)] + 110 \times 0.3522$$

$$= 76.5754$$

$$\text{IRR} \Rightarrow -76.5754 + 6 \times a_{10} + 110 v^{10} = 0$$

By trial and error $i = 10.5\%$

Since the IRR obtained is maximum under option 1, the investor should choose option 1

[10 marks]

Solution 8:

i) Let P be the annual premium

$$P \ddot{S}_{10} = 100000$$

$$P = 7,571.91$$

[2]

ii) a) Let the term of the stock purchased be n years

The discounted mean term at 5% p.a. is

$$\frac{(1 \times 3.125v) + (2 \times 3.125v^2) + \dots + (n \times 3.125v^n) + (n \times 100v^n)}{3.125v + 3.125v^2 + \dots + 3.125v^n + 100v^n} \text{ at 5\%}$$

$$= \frac{3.125 (Ia)_n + 100nv^n}{3.125a_n + 100v^n}$$

At $n = 12$, the DMT = 9.98

Hence the stock purchased has outstanding term of 12 years.

[4]

b) The amount invested was 7,571.92

The market price of the stock purchased for 100 nominal is

$$3.125a_{12} + 100v^{12} = 83.38 \text{ at 5\%}$$

The nominal amount purchased was

$$\frac{7571.92}{83.38} \times 100 = 9,081$$

[2]

[8 marks]

Solution 9:

$$E(i) = j = 0.08$$

$$S = [\text{var}(i)]^{1/2} = 0.07$$

i) The single premium accumulation has expected value

$$1000 E(s_{15}) = 1000 (1.08)^{15} = 3172.17$$

And standard deviation

$$1000 [\text{var}(s_{15})]^{1/2} = 1000 * ((1 + 2j + j^2 + s^2)^{15} - (1+j)^{30})^{0.5} = 808.14$$

[4]

ii) Assume that each year $\log(1+i_t)$ has Normal distribution.

$$\mu = 0.075$$

$$\sigma^2 = 0.065$$

$\log(s_{15})$ has a Normal distribution with mean $\mu' = 15\mu$ and standard deviation of $\sigma' = \sigma\sqrt{15}$

$$\mu' = 1.125 \text{ and } \sigma' = 0.987421$$

a) probability $(s_{15} < 0.6 E[s_{15}]) = \Phi\left[\frac{0.64359 - \mu'}{\sigma'}\right]$

$$= \Phi(-0.48754)$$

$$= 0.312937$$

[2]

$$\begin{aligned}
 \text{b) probability } (s_{15} > 1.5 E[s_{15}]) &= 1 - \Phi\left[\frac{1.55988 - \mu'}{\sigma'}\right] \\
 &= 1 - \Phi(0.440421) \\
 &= 1 - 0.670184 \\
 &= 0.329816
 \end{aligned}$$

[2]

[8 marks]

Solution 10:

i) An arbitrage opportunity exist if

- An investor can make a deal that would give him an immediate profit with no risk of future loss
- An investor can make a deal that has zero initial cost, no risk of future loss and a nonzero probability of a future profit

[2]

ii) Present value of dividends, I, is

$$I = 2(e^{-0.07 \times 0.25} + e^{-0.07 \times 0.5})$$

$$= 3.8965$$

$$T-t = \frac{7}{12} = 0.58333$$

Hence, forward price, F, is

$$F = (60 - 3.8965) e^{0.07 \times 0.58333} = 58.44$$

[4]

[6 marks]

Solution 11:

i) Present value is:

$$\sum_{t=1}^4 (1000 + 100t) v^{5t} \text{ at } 5\%$$

$$= 2751.54$$

Discounted mean term:

$$\sum_{t=1}^4 5t (1000 + 100t) v^{5t} / 2751.54 \quad \text{at } 5\%$$

$$= \frac{31609}{2751.54} = 11.5 \text{ years}$$

[4]

ii) Let amount invested in 10 year stock and 30 year stock be A and B respectively

$$A + B = 2751.54$$

$$\text{Value of 10 year stock} = V1 = A (0.05 a_{10+} v^{10})$$

The Discounted mean term

$$\frac{0.05(Ia)_{10} + 10v^{10}}{0.05 a_{10+} v^{10}} = T1 / (0.05 a_{10+} v^{10})$$

At interest rate of 5%

$V1 = A$ and $T1 = \ddot{a}_{10}$ at 5% = 8.1078

Similarly for the 30 year stock present value $v2 = B$ and the Discounted mean term $T2 = \ddot{a}_{30}$ at 5% = 16.141

Discounted mean term of the total asset =

$$\frac{(A \times 8.1078) + (B \times 16.141)}{A+B} = 16.141 - 0.00292A \text{ since } A+B = 2751.54$$

Equating the expression to 11.5

$A = 1593.87$ and $B = 1157.67$

[6]

[10 marks]
